

Chapter 14: Acid-Base Equilibrium

General Chemistry 2

Chem120

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Before we begin....

Topics you need to know :

Chemistry:

- 1) Equilibrium (Review Sewall Section Notes; Chapter 13) :
 - Write K (eq. constant) for a given reaction
 - Determine the value of K from given concentrations
 - ICE Tables
 - Effects of changing the conditions of the reaction on the system: Le Chatelier's Principle
- 2) Stoichiometry – balance equations!
- 3) Concentration calculations – determine moles/molarity/concentration of a solution/components

Math:

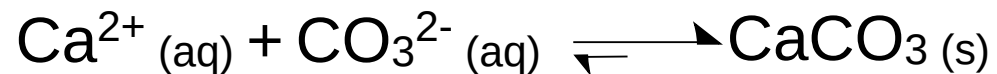
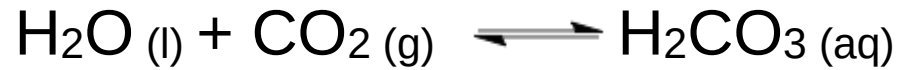
- 1) Quadratic Equations – solving quadratic equations (for ICE Table)
- 2) Log functions – Basic principles about logarithmic functions

By the end of this chapter you will be able to:

- Designate and differentiate acids/bases based on different theories
- Explain factors that affect acid/base strength
- Determine conjugate acids/bases
- Compare and contrast acid strengths
- Determine the relative acidity of salts
- Understand and solve acid-base equilibrium problems

But why do you need to know this?

30% of environmental CO_2 (g) is taken up by the oceans



Sea butterfly

It's all about acid-base equilibria!



Lab experiment: Sea butterfly shell in seawater pH ~ 7.8 over 45 days

Ocean surface water acidity

	pH	
1885	~ 8.2	Increasing acidity 
2005	~ 8.1	
2094	~ 7.8	

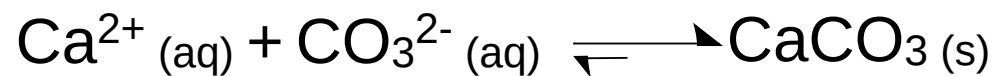


Image: National Oceanic and Atmospheric Administration

How are we going to do this?

Define acids and bases – these definitions will vary - there are different acid/base theories

- a) What are conjugate acids/bases?

Concept Video_1

Role of the solvent in acid/base equilibria – our solvent of choice – water

- a) What is K_w and what is pH?
- b) Compare relative acidity of solutions
- c) Calculating and measuring pH of solutions

Concept Video_2

Acid/base dissociation/association – Equilibria

- a) Relation between an acid(base) and its conjugate base(acid)
- b) K_a and pK_a
- c) Solving acid-base equilibria problem – ICE equations

Concept Video_3

Structure-function relation: Molecular properties and Acid strength

Acid-base properties of salt solutions

Concept Video_4

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Structure-function relation: Molecular properties and Acid strength

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Concept Video _4

What is an acid/base? **Arrhenius Theory**

Acid: Dissociates in water to form H_3O^+ . Proton donor



Base: Dissociates in water to form OH^- . Hydroxide ion donor

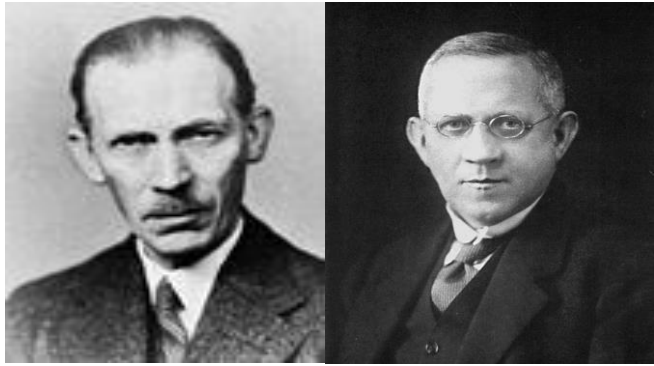


An Arrhenius **base** is a substance with OH in its formula that dissociates to yield OH^- .

This is one of earliest acid-base definition, which classifies these substances in terms of their behaviour in **water**.

More important in terms of historic understanding of acids-bases, but we won't be using this classification, as it is limited in how bases are defined.

What is an acid/base? Brønsted-Lowry theory



Acid: Proton (H^+) donor (Same definition as Arrhenius).



Base: Proton (H^+) acceptor.



What is an acid/base *reaction*? Brønsted-Lowry theory

Acid



- **Proton donor**, any species that donates an ***H⁺ ion***.

Bronsted-Lowry acid must **contain H in its formula**.

Base



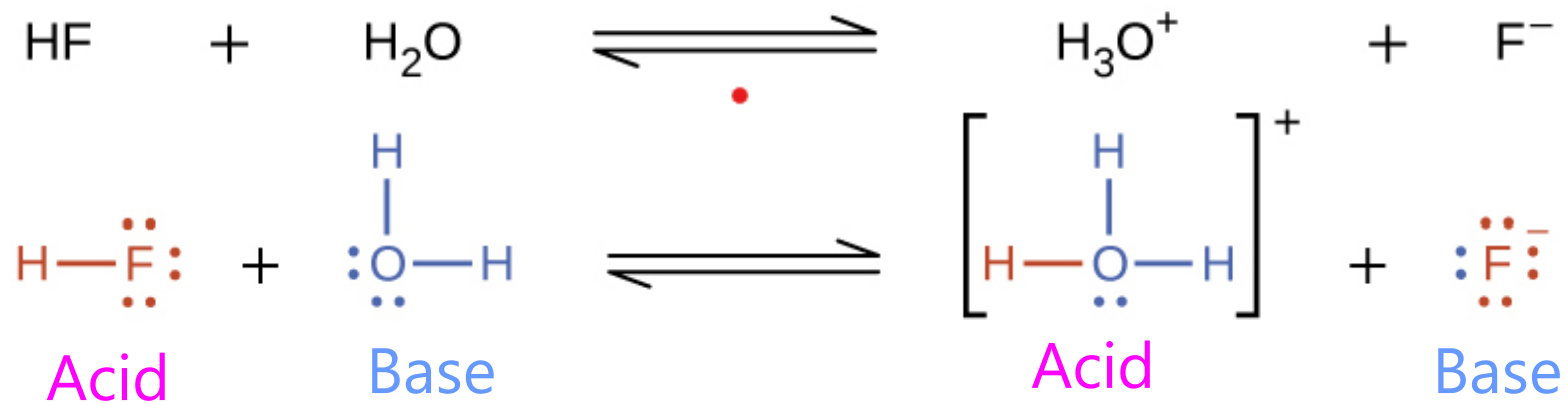
- **Proton acceptor**, any species that accepts an ***H⁺ ion***.

Bronsted-Lowry base must contain a **lone pair of electrons** to bond to H⁺.

An acid-base reaction is a ***proton-transfer process***.

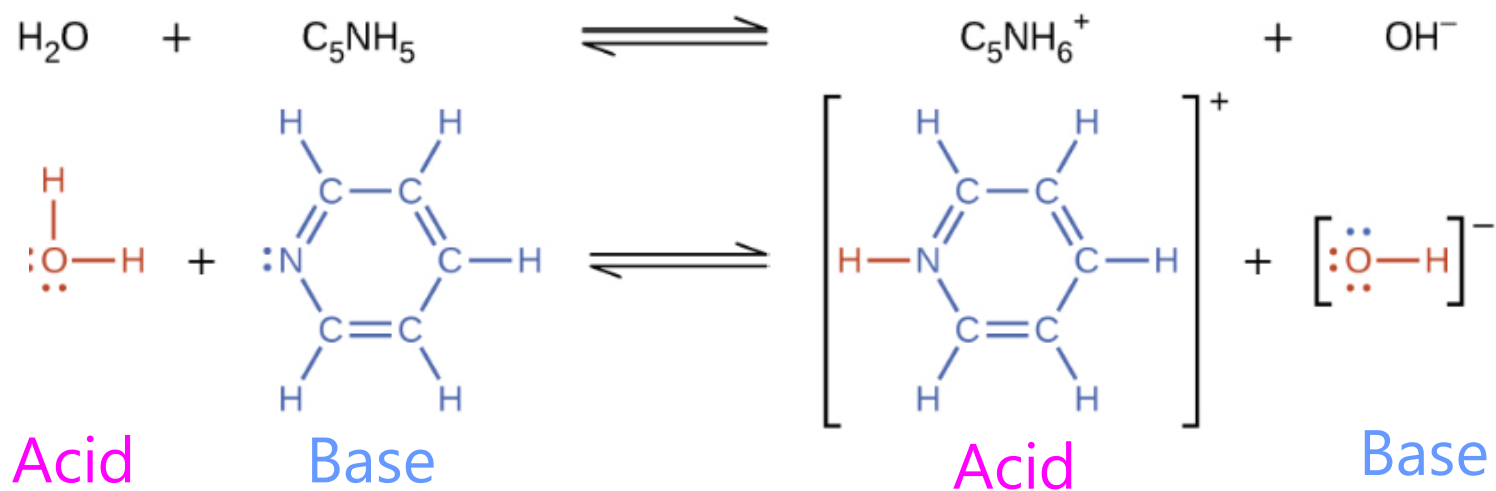
In a Lowry-Bronsted acid-base reaction, reactants include an acid and a base, and their conjugate base and acid are in the products, and vice versa

Acid/Base Equilibria



The proton exchange reaction between an **acid** (donates proton) and **water** is called **acid ionization**.

Water is AMPHIPROTIC – can react as either an acid or a base



The proton exchange reaction between a **base (accepts proton)** and **water** is called **base ionization**.

How are we going to do this?

Define acids and bases – these definitions will vary - there are different acid/base theories

- a) What are conjugate acids/bases?
- b) Relative acid/bases strength

Concept Video 1

Role of the solvent in acid/base equilibria – our solvent of choice – water

- a) What is K_w and what is pH?
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Concept Video 2

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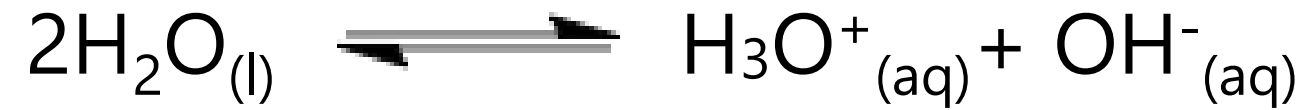
Concept Video 3

Structure-function relation: Molecular properties and Acid strength

Acid-base properties of salt solutions

Concept Video 4

Autoionization of Water



$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] \quad \text{We can measure } K_w!$$

This reaction is the auto-ionization of water (specifically, it is gaining and losing a proton by water) and since this is an equilibria, we have an associated equilibrium constant, which we call K_w

If we disturb the equilibria, by adding a base or an acid – the system responds (Le Chatelier's Principle) : K_w is a constant so increase in $[\text{H}_3\text{O}^+]/[\text{OH}^-]$ results in the decrease of the $[\text{OH}^-]/[\text{H}_3\text{O}^+]$ and we can measure this change

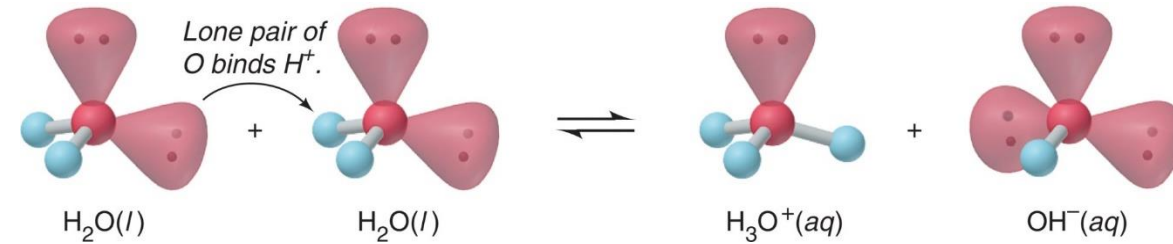
Higher $[\text{H}_3\text{O}^+]$ $\longrightarrow$ lower $[\text{OH}^-]$

Higher $[\text{OH}^-]$ $\longrightarrow$ lower $[\text{H}_3\text{O}^+]$

We can define the terms "acidic" and "basic" in terms of the relative concentrations of H_3O^+ and OH^- ions:

In an acidic solution,	$[\text{H}_3\text{O}^+] > [\text{OH}^-]$
In a neutral solution,	$[\text{H}_3\text{O}^+] = [\text{OH}^-]$
In basic solution,	$[\text{H}_3\text{O}^+] < [\text{OH}^-]$

Autoionization of Water



$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

The product: $[\text{H}_3\text{O}^+][\text{OH}^-]$ is a constant for a particular temperature

Think about why temperature is relevant here?

In pure water at 25°C ,

$$[\text{H}_3\text{O}^+] = [\text{OH}^-] = \sqrt{1.00 \times 10^{-14}} = 1.0 \times 10^{-7}$$

Both H_3O^+ and OH^- ions are present in aqueous solutions. Their relative concentrations determine whether an aqueous solution is acidic or basic

pH Scale: Measure of acidity/basicity

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$$

Take the log of the above equation

$$\log K_w = \log[\text{H}_3\text{O}^+] + \log[\text{OH}^-]$$

$$-\log K_w = -\log[\text{H}_3\text{O}^+] - \log[\text{OH}^-]$$

$$\text{p}K_w = \text{pH} + \text{pOH}$$

At 25 °C :

$$\text{p}K_w = -\log (1.00 \times 10^{-14}) = 14.00$$

$$\text{p}X = -\log[X]$$

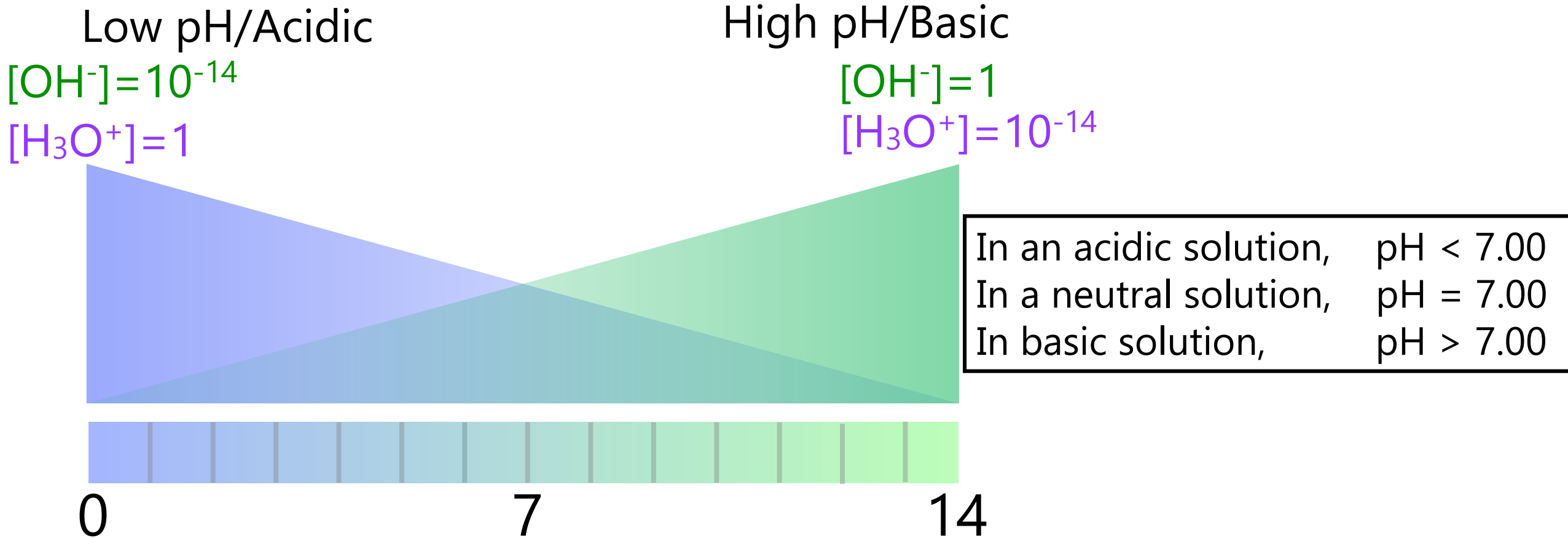
So:

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

$$14.00 = \text{pH} + \text{pOH}$$

pH scale: Measure of acidity/basicity

The **higher** the pH, the **lower** the $[\text{H}_3\text{O}^+]$ and the **less acidic** the solution.



pH Scale

[H ₃ O ⁺] (M)	[OH ⁻] (M)	pH	pOH	Sample Solution
10 ¹	10 ⁻¹⁵	-1	15	
10 ⁰ or 1	10 ⁻¹⁴	0	14	← 1 M HCl
10 ⁻¹	10 ⁻¹³	1	13	
10 ⁻²	10 ⁻¹²	2	12	← gastric juice ← lime juice
10 ⁻³	10 ⁻¹¹	3	11	← 1 M CH ₃ CO ₂ H (vinegar) ← stomach acid
10 ⁻⁴	10 ⁻¹⁰	4	10	← wine ← orange juice
10 ⁻⁵	10 ⁻⁹	5	9	← coffee
10 ⁻⁶	10 ⁻⁸	6	8	← rain water
10 ⁻⁷	10 ⁻⁷	7	7	← pure water
10 ⁻⁸	10 ⁻⁶	8	6	← blood
10 ⁻⁹	10 ⁻⁵	9	5	← ocean water ← baking soda
10 ⁻¹⁰	10 ⁻⁴	10	4	
10 ⁻¹¹	10 ⁻³	11	3	← Milk of Magnesia
10 ⁻¹²	10 ⁻²	12	2	← household ammonia, NH ₃
10 ⁻¹³	10 ⁻¹	13	1	← bleach
10 ⁻¹⁴	10 ⁰ or 1	14	0	← 1 M NaOH
10 ⁻¹⁵	10 ¹	15	-1	

At 25 °C; $K_w = 1.0 \times 10^{-14}$

$$1.00 \times 10^{-14} = [\text{H}_3\text{O}^+][\text{OH}^-]$$

Take -log of the above equation:

$$14.00 = \text{pH} + \text{pOH}$$

Since K_w is a constant, the values of pH, pOH, [H₃O⁺], and [OH⁻] are interrelated:

If [H₃O⁺] increases, [OH⁻] decreases (and vice versa).

If pH increases, pOH decreases (and vice versa).

Measuring pH

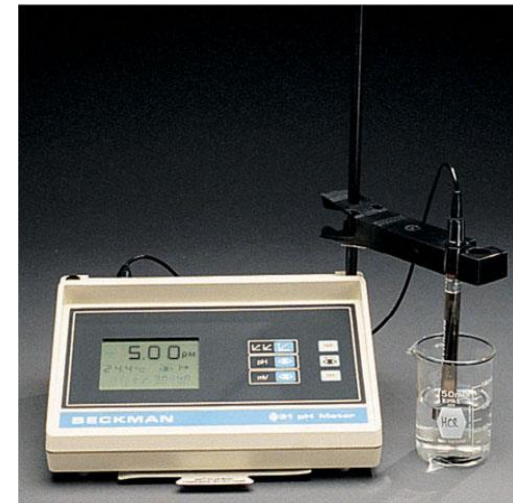


pH paper

Red litmus paper with a drop of base here



Blue litmus paper with a drop of acid here



pH meter

How are we going to do this?

Define acids and bases – these definitions will vary - there are different acid/base theories

- a) What are conjugate acids/bases?
- b) Relative acid/bases strength

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Acidic/Basic Strength

What determines acidic/basic strength?

Can you quantify (calculate) acidic/basic strength?

Applying principles from Equilibria

We are going to apply the same principles you learned with Prof. Sewall in “Chapter 13: Equilibrium” to a system where the reactants/products are acids/bases



In this examples:

HA is an acid

B is a base

What does it mean if the **equilibrium constant, K, for the above reaction is $\gg 1$**

$K \gg 1$ implies that the **forward reaction is favorable** i.e. at equilibrium we will have large quantities of the products HB^+ and A^-

Think of this in terms of acids and bases, a large K implies:

- Most of the acid, HA will ionize to form its conjugate base, A^- .
- Most of the base, B will ionize to form its conjugate acid, HB^+ .

We will come back to this idea in a few slides

Applying principles from Equilibria

We are going to apply the same principles you learned with Prof. Sewall in “Chapter 13: Equilibrium” to a system where the reactants/products are acids/bases



In this example:

HA is an acid (with A^- as its conjugate base)

B is a base (with HB^+ as its conjugate acid)

To be able to compare acids and bases, we need some sort of a reference.

We measure the **equilibrium constant of the proton-exchange reaction of the acid (or the base) with water.**

This helps us compare different acids and bases to each other because we have a reference.

Acidic Strength

Adding acid to water – Has an associated equilibrium constant (K)

Here:
HA is the acid
H₂O is the base



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

Acid dissociation constant: K_a
This defines the relative strength of the acid compared to water

Acidic Strength: Strong Acids



A **strong** acid dissociates **completely** into ions in water K_a is irrelevant for strong acids (there is no equilibria)

An aqueous solution of a **strong** acid contains **no HA molecules** (all molecules have dissociated to give A^-)

For a **strong acid**, we can expect the value of $K_a > 1$ (tough to measure this value experimentally)

All strong acids are equally strong in water.

For example: Aqueous solutions of 1M HCl and 1M HNO_3 have the same $[\text{H}_3\text{O}^+]/\text{pH}$

Strong Acids

6 Strong Acids	
HClO_4	perchloric acid
HCl	hydrochloric acid
HBr	hydrobromic acid
HI	hydroiodic acid
HNO_3	nitric acid
H_2SO_4	sulfuric acid

You should know the name and chemical formula of the most common strong acids

All strong acids are equally strong in water.

For example: Aqueous solutions of 1M HCl and 1M HNO_3 have the same $[\text{H}_3\text{O}^+]/\text{pH}$

Relative Acidic Strength: Comparing K_a

Acid-base equilibria - associated equilibrium constant (K)



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

K_a is an important quantity for weak acids (where there is an equilibrium)

The value of K_a is an indication of acid strength (directly depends on $[\text{H}_3\text{O}^+]$) – helps us compare acids to each other

Stronger acid $\longrightarrow$ **higher $[\text{H}_3\text{O}^+]$** $\longrightarrow$ **larger K_a**

Weaker acid $\longrightarrow$ **lower % dissociation of HA** $\longrightarrow$ **smaller K_a**

Relative Acidic Strength: Comparing K_a

Name (Formula)	Lewis Structure*	K_a
Chlorous acid (HClO_2)	$\text{H}-\ddot{\text{O}}-\ddot{\text{Cl}}=\ddot{\text{O}}$	1.1×10^{-2}
Nitrous acid (HNO_2)	$\text{H}-\ddot{\text{O}}-\ddot{\text{N}}=\ddot{\text{O}}$	7.1×10^{-4}
Hydrofluoric acid (HF)	$\text{H}-\ddot{\text{F}}:$	6.8×10^{-4}
Formic acid (HCOOH)	$\begin{array}{c} \text{:O:} \\ \parallel \\ \text{H}-\text{C}-\ddot{\text{O}}-\text{H} \end{array}$	1.8×10^{-4}
Acetic acid (CH_3COOH)	$\begin{array}{c} \text{H} \quad \text{:O:} \\ \quad \parallel \\ \text{H}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \\ \text{H} \end{array}$	1.8×10^{-5}
Propanoic acid ($\text{CH}_3\text{CH}_2\text{COOH}$)	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{:O:} \\ \quad \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	1.3×10^{-5}
Hypochlorous acid (HClO)	$\text{H}-\ddot{\text{O}}-\ddot{\text{Cl}}:$	2.9×10^{-8}
Hydrocyanic acid (HCN)	$\text{H}-\text{C}\equiv\text{N}:$	6.2×10^{-10}

*Red type indicates the ionizable proton; all atoms have zero formal charge.

Strong acids include:

Hydrohalic acids (**HCl, HBr, and HI**) but NOT HF.

Oxoacids in which the number of O atoms exceeds the number of ionizable protons by two or more (eg., **HNO₃, H₂SO₄, HClO₄**.)

ACID STRENGTH

Relation between K_a and pK_a

Table 16.5 The Relationship between K_a and pK_a

Acid Name (Formula)	K_a at 25°C	pK_a
Hydrogen sulfate ion (HSO_4^-)	1.0×10^{-2}	1.99
Nitrous acid (HNO_2)	7.1×10^{-4}	3.15
Acetic acid (CH_3COOH)	1.8×10^{-5}	4.75
Hypobromous acid (HBrO)	2.3×10^{-9}	8.64
Phenol ($\text{C}_6\text{H}_5\text{OH}$)	1.0×10^{-10}	10.00

We will learn
more about
 pK_a and its
relevance next
week

$$pK_a = -\log K_a$$

A low pK_a corresponds to a high K_a .

Basic Strength

Adding a base to water – Must have an associated equilibrium constant (K)

Here:

B is the base

H₂O is the acid



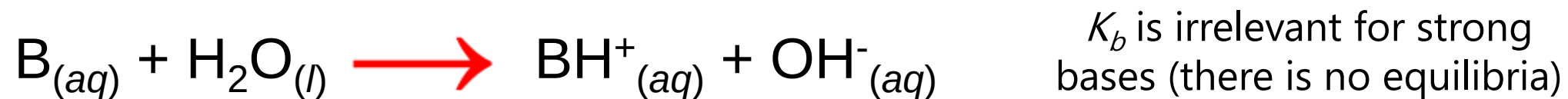
Base Ionization constant: K_b

This defines the relative strength of the base compared to water

$$K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$$

Basic Strength: Strong Base

Acid-base equilibria - associated equilibrium constant (K)



What is a strong base? A **strong** base **completely** ionizes to produce OH^-

An aqueous solution of a **strong** base contains **no B molecules**.

For a **strong base**, we can expect the value of $K_b > 1$ (tough to measure this value experimentally)

All strong bases are equally strong in water.

For example: Aqueous solutions of 1M NaOH and 1M KOH have the same $[\text{H}_3\text{O}^+]/\text{pH}$

Strong Bases

6 Strong Bases	
LiOH	lithium hydroxide
NaOH	sodium hydroxide
KOH	potassium hydroxide
Ca(OH) ₂	calcium hydroxide
Sr(OH) ₂	strontium hydroxide
Ba(OH) ₂	barium hydroxide

You should know the name and chemical formula of the most common strong bases

All strong bases are equally strong in water.
For example: Aqueous solutions of 1M NaOH and 1M KOH have the same $[\text{H}_3\text{O}^+]/\text{pH}$

Basic Strength: Weak Base

Acid-base equilibria - associated equilibrium constant (K)



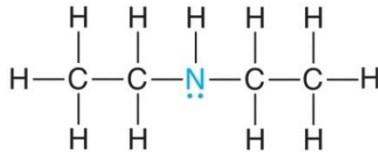
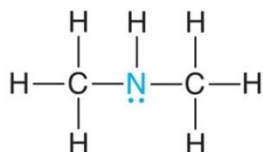
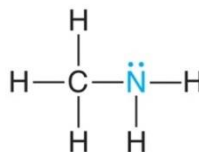
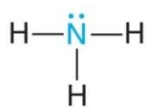
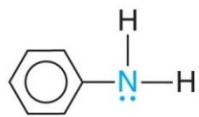
$$K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$$

Base ionization constant for a weak base has a small value

A **weak** base forms a few OH^- ions in water.

A dilute solution of a **weak** base has **mostly B** molecules.

Relative Basic Strength: Comparing K_b

Name (Formula)	Lewis Structure*	K_b
Diethylamine $[(CH_3CH_2)_2NH]$		8.6×10^{-4}
Dimethylamine $[(CH_3)_2NH]$		5.9×10^{-4}
Methylamine (CH_3NH_2)		4.4×10^{-4}
Ammonia (NH_3)		1.76×10^{-5}
Pyridine (C_5H_5N)		1.7×10^{-9}
Aniline $(C_6H_5NH_2)$		4.0×10^{-10}

*Blue type indicates the basic nitrogen and its lone pair.

BASIC STRENGTH

Strong bases include

water-soluble compounds containing O^{2-} or OH^- ions.

M_2O or MOH , where M = Group 1 metal (Li, Na, K, Rb, Cs)

MO or $M(OH)_2$ where M = group 2 metal (Ca, Sr, Ba).

Weak bases include

ammonia (NH_3),

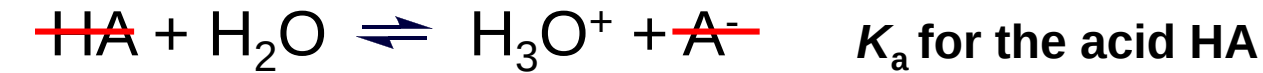
amines, which have the general formula: RNH_2 or R_2NH

Commonly: N atom with a lone electron pair.

Relation between Acids (or Bases) and Their Conjugates

A⁻ is the base

HA is its conjugate acid



K_c for the overall equation = $K_a \times K_b$, so

$$\frac{[\text{H}_3\text{O}^+]\cancel{[\text{A}^-}]}{\cancel{[\text{HA}]}} \times \frac{\cancel{[\text{HA}]}\text{[OH}^-]}{\cancel{[\text{A}^-}]} = [\text{H}_3\text{O}^+][\text{OH}^-]$$

$$K_a \quad \times \quad K_b \quad = \quad K_w$$

This relationship is true for any ***conjugate acid-base pair*** (in aqueous media)

Acid			Base		
Increasing acid strength ↑	perchloric acid	HClO ₄	Do not undergo base ionization in water	ClO ₄ ⁻	perchlorate ion
	sulfuric acid	H ₂ SO ₄		HSO ₄ ⁻	hydrogen sulfate ion
	hydrogen iodide	HI		I ⁻	iodide ion
	hydrogen bromide	HBr		Br ⁻	bromide ion
	hydrogen chloride	HCl		Cl ⁻	chloride ion
	nitric acid	HNO ₃		NO ₃ ⁻	nitrate ion
	hydronium ion	H ₃ O ⁺		H ₂ O	water
	hydrogen sulfate ion	HSO ₄ ⁻		SO ₄ ²⁻	sulfate ion
	phosphoric acid	H ₃ PO ₄		H ₂ PO ₄ ⁻	dihydrogen phosphate ion
	hydrogen fluoride	HF		F ⁻	fluoride ion
	nitrous acid	HNO ₂		NO ₂ ⁻	nitrite ion
	acetic acid	CH ₃ CO ₂ H		CH ₃ CO ₂ ⁻	acetate ion
	carbonic acid	H ₂ CO ₃		HCO ₃ ⁻	hydrogen carbonate ion
	hydrogen sulfide	H ₂ S		HS ⁻	hydrogen sulfide ion
	ammonium ion	NH ₄ ⁺		NH ₃	ammonia
	hydrogen cyanide	HCN		CN ⁻	cyanide ion
	hydrogen carbonate ion	HCO ₃ ⁻		CO ₃ ²⁻	carbonate ion
	water	H ₂ O		OH ⁻	hydroxide ion
	hydrogen sulfide ion	HS ⁻	Undergo complete base ionization in water	S ²⁻	sulfide ion
	ethanol	C ₂ H ₅ OH		C ₂ H ₅ O ⁻	ethoxide ion
	ammonia	NH ₃		NH ₂ ⁻	amide ion
	hydrogen	H ₂		H ⁻	hydride ion
	methane	CH ₄		CH ₃ ⁻	methide ion

$$K_a \times K_b = K_w$$

For the acid For the base Constant

If we know K_a of an acid (or K_b of a base), we can calculate the K_b of the *conjugate base* (or K_a of the *conjugate acid*)

The *stronger* the acid is, the *weaker* its conjugate base.

Figure 14.8 This figure shows strengths of conjugate acid-base pairs relative to the strength of water as the reference substance.

What can we learn from conjugate acid/bases?

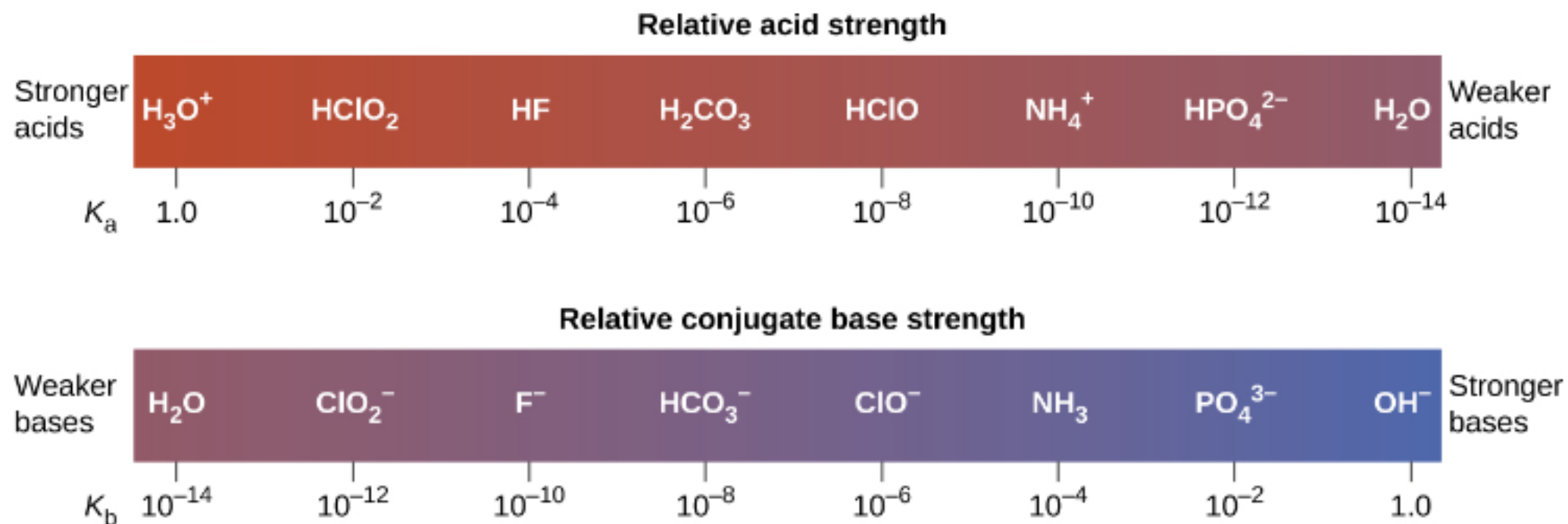


Figure 14.7 Relative strengths of several conjugate acid-base pairs are shown.

If you know the K_a of an acid – you can determine the K_b of the conjugate base
If you know the K_b of a base – you can determine the K_a of the conjugate acid.

How to determine the pH of an aqueous solution of an acid or base?

We are going to learn how to do this through practice.

There are two main scenarios:

1. Is it a **strong** acid/base? (100% dissociated – no equilibria)
2. Is it a **weak** acid/base? (We will use the relevant K_a or K_b for the acid or base)

Example 1: What is the pH of 0.1 M NaOH at 298 K?

Think about what is being asked – is it an acid or a base? Is it a strong acid or a base:

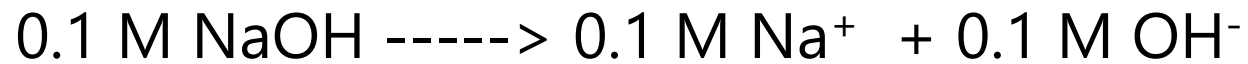
Strong base: NaOH

The concentration of the strong base is much higher than that of the only other base present in the aqueous solution (water)

Assumptions you can make:

- Strong base or acid is 100% dissociated.
- Strong base or acid is the **only** source of $[\text{OH}^-]$ or $[\text{H}_3\text{O}^+]$

Example 1: What is the pH of 0.1 M NaOH at 298 K?



$$[\text{Base}] = [\text{OH}^-] = 0.1 \text{ M}$$

We can now calculate $[\text{H}_3\text{O}^+]$

$$[\text{H}_3\text{O}^+] = K_w / [\text{OH}^-] = 10^{-14} / 0.1 = 10^{-13}$$

$$\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log (10^{-13}) = 13$$

The final answer makes sense - we expect this to be a highly basic solution

There is another base present (water) – but we can ignore it because the contribution from water to $[\text{OH}^-]$ is negligible compared to 0.1 M

Example 2: What is the pH of a 10^{-8} M HNO_3 at 298 K ?

Let's try to solve the problem like we did for example 1

$$[\text{Acid}] = [\text{H}_3\text{O}^+] = 10^{-8} \text{ M}$$

Now calculate pH

$$[\text{H}_3\text{O}^+] = 10^{-8} \text{ M}$$

This does not make sense - we expect this to be an acidic solution (we have added an acid to water, even if it is a small amount, we expect the acid to decrease the pH of neutral water and not increase it!)

Example 2: What is the pH of a 10^{-8} M HNO_3 at 298 K ?

We cannot solve a problem like this like we did for example 1 because here the acid is **dilute**

What does **dilute** mean in for strong acids/bases?

Concentration $< 10^{-6}$ M

- Strong base or acid is 100% dissociated.
- Strong base or acid is **NOT** the only source of $[\text{OH}^-]$ or $[\text{H}_3\text{O}^+]$

You can try to do this on your own – we will practice this in class

Example 3:

What is the pH of vinegar?

Vinegar is a 0.8 M solution of acetic acid CH_3COOH ($K_a = 1.74 \times 10^{-5}$)?

This is an example of a weak acid – How do we know that?

We have a K_a value given!

We are going to **use ICE** (like in Chapter 13 – equilibrium) to solve this

What is the chemical equation that we would use here for the ICE table?

Hint: What is the equilibrium constant given? Which chemical equation does that represent?

You can try to do this on your own – we will practice this in class

Example 3:

What is the pH of vinegar?

Vinegar is a 0.8 M solution of acetic acid CH_3COOH ($K_a = 1.74 \times 10^{-5}$)?

Example 3 is an example of a concentrated weak acid/base

In general, for a concentrated solution of a weak acid, $[HA] \gg K_a$

$$[H_3O^+] = \sqrt{[HA]_{\text{initial}} K_a}$$

In general, for a concentrated solution of a weak base, $[BOH] \gg K_b$

$$[OH^-] = \sqrt{[BOH]_{\text{initial}} K_b}$$

Example 4:

What is the pH of 8.00×10^{-5} M solution of vinegar (acetic acid)?
($K_a = 1.74 \times 10^{-5}$)?

This is an example of a weak acid - How do we know that?

We have a K_a value given!

We are going to **use ICE** (like in Chapter 13 – equilibrium) to solve this

What is the chemical equation that we would use here for the ICE table?

Hint: What is the equilibrium constant given? Which chemical equation does that represent?

You can try to do this on your own – we will
practice this in class

How are we going to do this?

Define acids and bases – these definitions will vary - there are different acid/base theories

- a) What are conjugate acids/bases?
- b) Relative acid/bases strength

Concept Video_1

Role of the solvent in acid/base equilibria – our solvent of choice – water

- a) What is K_w and what is pH?
- b) Compare relative acidity of solutions
- c) Calculating and measuring pH of solutions

Concept Video_2

Acid/base dissociation/association – Equilibria

- a) Relation between an acid(base) and its conjugate base(acid)
- b) K_a and pK_a
- c) Solving acid-base equilibria problem – ICE equations

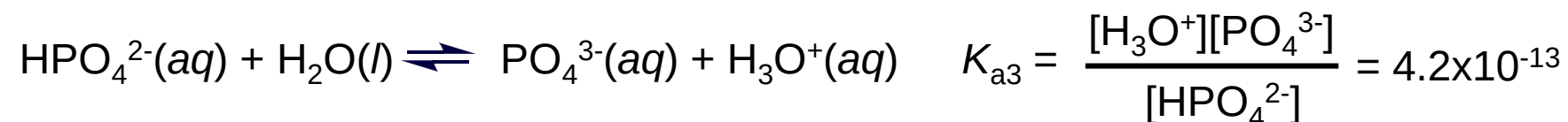
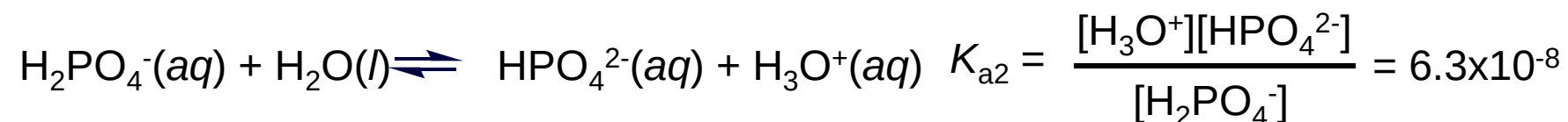
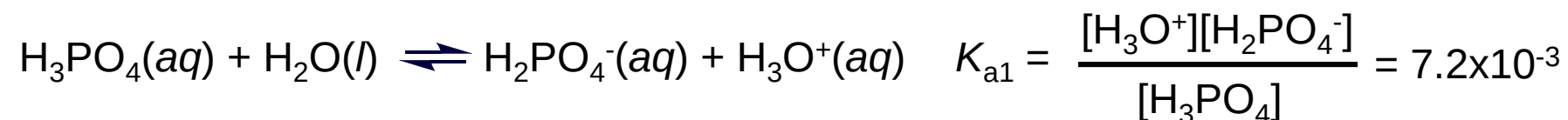
Concept Video_3

Structure-function relation: Molecular properties and Acid strength
Acid-base properties of salt solutions

Concept Video_4

Polyprotic Acids

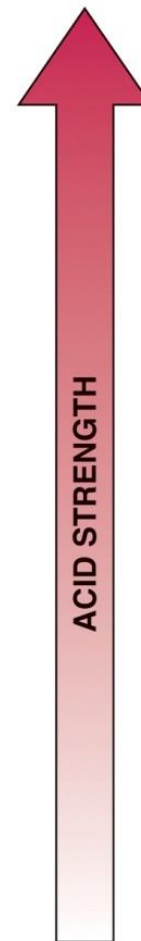
A **polyprotic acid** is an acid with more than one ionizable proton. In solution, each dissociation step has a different value for K_a :



$$K_{a1} > K_{a2} > K_{a3}$$

Polyprotic Acids

Name (Formula)	Lewis Structure*	K_{a1}	K_{a2}	K_{a3}
Oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$)	$\begin{array}{c} \text{:O:} \quad \text{:O:} \\ \parallel \quad \parallel \\ \text{H}-\ddot{\text{O}}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \end{array}$	5.6×10^{-2}	5.4×10^{-5}	
Sulfurous acid (H_2SO_3)	$\begin{array}{c} \text{:O:} \\ \parallel \\ \text{H}-\ddot{\text{O}}-\ddot{\text{S}}-\ddot{\text{O}}-\text{H} \end{array}$	1.4×10^{-2}	6.5×10^{-8}	
Phosphoric acid (H_3PO_4)	$\begin{array}{c} \text{:O:} \\ \parallel \\ \text{H}-\ddot{\text{O}}-\text{P}-\ddot{\text{O}}-\text{H} \\ \\ \text{:O}-\text{H} \end{array}$	7.2×10^{-3}	6.3×10^{-8}	4.2×10^{-13}
Arsenic acid (H_3AsO_4)	$\begin{array}{c} \text{:O:} \\ \parallel \\ \text{H}-\ddot{\text{O}}-\text{As}-\ddot{\text{O}}-\text{H} \\ \\ \text{:O}-\text{H} \end{array}$	6×10^{-3}	1.1×10^{-7}	3×10^{-12}
Carbonic acid (H_2CO_3)	$\begin{array}{c} \text{:O:} \\ \parallel \\ \text{H}-\ddot{\text{O}}-\text{C}-\ddot{\text{O}}-\text{H} \end{array}$	4.5×10^{-7}	4.7×10^{-11}	
Hydrosulfuric acid (H_2S)	$\text{H}-\ddot{\text{S}}-\text{H}$	9×10^{-8}	1×10^{-17}	



*Red type indicates the ionizable protons.

Molecular Properties and Acidic Strength

For nonmetal hydrides, acid strength depends on:

- 1) the electronegativity of the central nonmetal, and
- 2) the strength of the bond.

Across a period, acid strength ***increases***.

Electronegativity increases across a period, so the acidity of increases.

Down a group, acid strength ***increases***.

The length of the bond increases down a group and its bond strength therefore decreases.

14	15	16	17	
6 CH₄ Neither acid nor base	7 NH₃ Weak base $K_b = 1.8 \times 10^{-5}$	8 H₂O Neutral	9 HF Weak acid $K_a = 6.8 \times 10^{-4}$	↑ Increasing base strength ↓ Increasing acid strength
14 SiH₄ Neither acid nor base	15 PH₃ Very weak base $K_b = 4 \times 10^{-28}$	16 H₂S Weak acid $K_a = 9.5 \times 10^{-8}$	17 HCl Strong acid	
→ Increasing acid strength				
← Increasing base strength				

Molecular Properties and Acidic Strength

Fluorine (F)	Chlorine (Cl)	Bromine (Br)	Iodine (I)
HF	HCl	HBr	HI
weakest acid			strongest acid

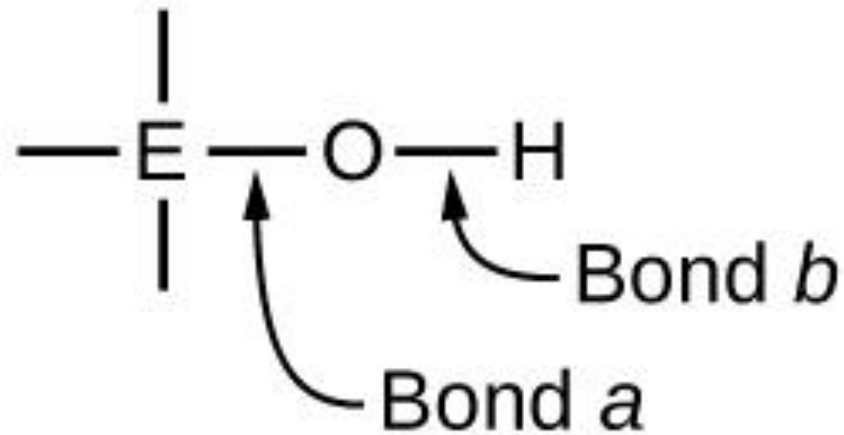
Increasing E-H bond lengths, increasing acidity



These are hydrides (not oxoacids) – compare bond lengths to determine acidity

Molecular Properties and Acidic Strength: Oxo Acids

- All oxoacids have the acidic H bonded to an ***O atom***.
- Acid strength of oxoacids depends on:
 - the electronegativity of the central nonmetal (E), and
 - the number of O atoms around the central nonmetal (E),

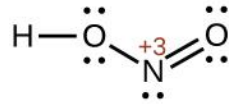


Molecular Properties and Acidic Strength: Oxo Acids

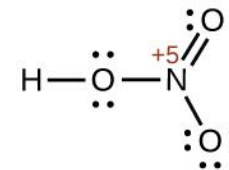
Comparing oxoacids:

- 1) For molecules with the **same number of oxygen atoms**: The more electronegative atom bound to the oxygen – the higher the acidity (Example: HOF is more acidic than HOCl or HOI)
- 2) For the **same central atom**: more oxygen – higher acidity (Example: HClO_3 is more acidic than HClO_2)

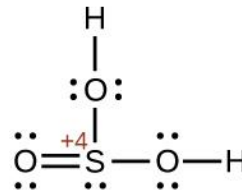
Nitrous acid



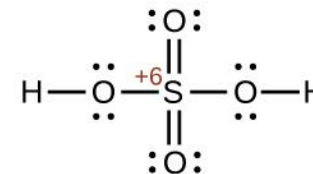
Nitric acid



Sulfurous acid



Sulfuric acid



Increasing acid strength



Acid-Base Properties of Salt Solutions

Some salts can affect the $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ in solution

You should be able to identify the cation/anion in the salt.

So, for example if you are given a salt of the type:

NaCl (or KCl or LiCl)

You should be able to identify that in water this will dissociate to form:

$\text{Na}^+ + \text{Cl}^-$

Now, you will look at the effect of each ion (cation and anion) on the concentrations of $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$

Acid-Base Properties of Salt Solutions

A solution of NaNO_3 will be acidic or basic or neutral?

NaNO_3 will form $\text{Na}^+(\text{aq})$ and $\text{NO}_3^-(\text{aq})$ ions in solution.

Both are non-bases/acids – will not change concentration of H_3O^+ or OH^-

So, the solution is neutral

A solution of NH_4Cl will be acidic or basic or neutral?

NH_4Cl will form $\text{NH}_4^+(\text{aq})$ and $\text{Cl}^-(\text{aq})$ ions in solution.

$\text{Cl}^-(\text{aq})$ is a non-base (conjugate of a very strong acid, HCl) and will not affect the pH.

NH_4^+ is a weak acid (conjugate of a weak base NH_3).

So, the solution is acidic